Criterion 4: Students' Performance (120)

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Admission Details for the Program (with focus on *excluding multiple entry/exit points*)

Purpose of this Table in NBA

- **To show student intake pattern and program demand over the last six academic years.**
- **Helps assessors evaluate:**
  - **Consistency of admissions (program reputation & demand).**
  - **Quality of intake (if sanctioned seats are filled).**
  - **Program stability (minimal dropouts or migrations).**

Table Structure (Required by NBA)

Item	CAY	CAYm1	CAYm2	CAYm3	CAYm4 (LYG)	CAYm5 (LYGm1)	CAYm6 (LYGm2)
N = Sanctioned Intake (AICTE/Competent Authority)							
N1 = 1st Year Admissions (adjusted for migration in/out)							
N2 = Lateral Entry Admissions (2nd year)							
N3 = Separate Division Admissions (if any)							
N4 = Supernumerary Admissions (TFWS/EWS/PMSSS, etc.)							
Total Admissions = N1 + N2 + N3 + N4							

Explanation of Each Parameter

1. **N = Sanctioned Intake**
   - Approved intake capacity as per AICTE/competent authority for each year.
   - Example: If B.Tech (CSE) has 120 seats sanctioned → N = 120.
   - If the intake changed (e.g., 60 → 120), reflect separately in each year's column.
2. **N1 = Total 1st Year Admissions**
   - Represents actual students admitted in Year 1.
   - Adjustments required:
     - Subtract students who migrated out to another program.
     - Add students who migrated in from another program.
   - Example:
     - 110 students admitted, 2 migrated out, 3 migrated in → N1 = 111.
3. **N2 = Lateral Entry Admissions**
   - Students admitted in 2nd year through diploma lateral entry scheme.
   - Also includes admissions to fill leftover seats from 1st year (if permitted).
   - Example: 10 diploma students admitted in 2nd year → N2 = 10.
4. **N3 = Separate Division Admissions (if applicable)**
   - Some colleges run parallel divisions for the same program code.
   - If yes, show separately here.
   - Example: If AICTE approved an additional batch of 60 in 2019, it should be recorded here.
5. **N4 = Supernumerary Admissions**
   - Admissions over and above sanctioned intake, approved under specific quotas:
     - TFWS (Tuition Fee Waiver Scheme)
     - EWS (Economically Weaker Section)
     - PMSSS (Prime Minister's Special Scholarship Scheme for J&K/NE students)
     - Foreign Nationals/PIOs (if AICTE approved)
   - These should be explicitly shown; not merged with N1/N2.
6. **Total Admissions = N1 + N2 + N3 + N4**
   - Gives the final number of students admitted to the program each academic year.
   - *Important*: Students admitted under Multiple Entry and Exit System (MEES/Academic Bank of Credits) are not included.
   - Reason: NBA considers only regular degree-seeking cohort data for outcome-based accreditation.

Time Labels (CAY, LYG, etc.)

CAY → Current Academic Year (the year of NBA assessment, e.g., 2024–25).
CAYm1 → Current Academic Year minus 1 (e.g., 2023–24).
CAYm2 → Current Academic Year minus 2 (e.g., 2022–23).
CAYm3 → Current Academic Year minus 3.
CAYm4 (LYG) → Last Year Graduated batch.
CAYm5 (LYGm1) → Graduated batch minus 1 year.
CAYm6 (LYGm2) → Graduated batch minus 2 years.

Evidence Required for NBA

- Sanctioned Intake Proof → AICTE approval letters for each year.
- Admission Data → University/State admission authority records.
- Migration Records → Transfers in/out approved by affiliating body.
- Lateral Entry Proof → Diploma entry admission list.
- Supernumerary Proof → AICTE notification, admission committee records.
- Exclusion Proof → Documentation showing no MEES students were included.

Item	CAY (2024-25)	CAYm1 (2023-24)	CAYm2 (2022-23)	CAYm3 (2021-22)	CAYm4 / LYG (2020-21)	CAYm5 / LYGm1 (2019-20)	CAYm6 / LYGm2 (2018-19)
N = Sanctioned Intake (AICTE/Authority)	120	120	120	120	120	120	120
N1 = 1st Year Admissions (adjusted for migration in/out)	111	118	109	115	102	95	101
N2 = Lateral Entry Admissions (2nd year)	8	5	10	7	12	14	9
N3 = Separate Division Admissions (if any)	0	0	0	0	0	0	0
N4 = Supernumerary Admissions (TFWS/EWS/PMSSS, etc.)	7	4	6	5	4	3	2
Total Admissions = N1 + N2 + N3 + N4	126	127	125	127	118	112	112

How to Interpret this Example

- **Seat Utilisation (Total ÷ N)**
  - **CAY = 126 ÷ 120 = 105% (extra due to supernumerary)**
  - **CAYm1 = 127 ÷ 120 = 106%**
  - **CAYm2 = 125 ÷ 120 = 104%**
  - **CAYm3 = 127 ÷ 120 = 106%**
  - **CAYm4 = 118 ÷ 120 = 98%**
  - **CAYm5 = 112 ÷ 120 = 93%**
  - **CAYm6 = 112 ÷ 120 = 93%**
- **Enrolment Ratio (NBA 4.1) for last 3 intakes (based on N1 only)**
  (111+118+109)÷(120+120+120)(111 + 118 + 109) ÷ (120 + 120 + 120)(111+118+109)÷(120+120+120) × 100 = 94.17%
- **Note on MEES/ABC students**
  - **Some batches had students through Multiple Entry–Exit System (Academic Bank of Credits).**
  - **These students were excluded from N1/N2/N4 and hence not shown in this table (as per NBA rules).**

Admission Details for the Program through Multiple Entry and Exit Points (N5 & N6 Tracking Table)

A. Objective

- **To record and monitor student movement under the Multiple Entry and Exit System (MEES) and Academic Bank of Credits (ABC), as per NEP-2020 guidelines.**
- **To ensure transparency in re-entries (N5) and exits (N6) within each batch, supported by official evidence such as ABC Ledgers, University Approvals, and Exit Certificates.**

B. Key Definitions

Code	Meaning	Description
N5 (Multiple Entry)	Total no. of students admitted in higher years (2nd / 3rd / 4th year) through re-entry, credit transfer, or lateral entry under MEES/ABC.	N5=N52+N53+N54N5 = N52 + N53 + N54N5=N52+N53+N54
N52	Students admitted in 2nd year via MEES/ABC	Example: Lateral entry after diploma or prior credits.
N53	Students admitted in 3rd year via MEES/ABC	Example: Credit re-entry after 2-year diploma.
N54	Students admitted in 4th year via MEES/ABC	Example: Returning learner completing B.Tech final year.
N6 (Multiple Exit)	Total no. of students exiting the program after 1st, 2nd, or 3rd year through MEES/ABC.	N6=N61+N62+N63N6 = N61 + N62 + N63N6=N61+N62+N63
N61	Students exiting after 1st year (Certificate Level).	Certificate awarded after 1 year of credits.
N62	Students exiting after 2nd year (Diploma Level).	Diploma in Engineering/Science.
N63	Students exiting after 3rd year (Advanced Diploma Level).	Advanced Diploma before full degree.

Admission and Exit Details through Multiple Entry and Exit Points

Item (No. of Students Admitted / Exited)	CAY (2024-25)	CAYm1 (2023-24)	CAYm2 (2022-23)	CAYm3 (2021-22)	CAYm4 (LYG 2020-21)	CAYm5 (LYGm1 2019-20)	CAYm6 (LYGm2 2018-19)
N52 – Students admitted in 2nd Year	3	4	2	2	1	0	0
N53 – Students admitted in 3rd Year	1	2	0	0	1	0	0
N54 – Students admitted in 4th Year	0	1	0	0	0	0	0
➞ N5 (Total Multiple Entry)	4	7	2	2	2	0	0
N61 – Students exiting after 1st Year	1	0	0	1	0	0	0
N62 – Students exiting after 2nd Year	1	2	1	0	0	0	0
N63 – Students exiting after 3rd Year	0	1	0	0	0	0	0
➞ N6 (Total Multiple Exit)	2	3	1	1	0	0	0

No. of Students Graduated Within the Stipulated Period of the Program

A. Objective

- To record the graduation efficiency of each batch—how many students completed the program within the normal duration (4 years for UG engineering).
- This directly contributes to the Program Outcome (PO12 – Lifelong Learning) and the Program Educational Objective (PEO) on timely academic progression.

B. Key Terms and Formulae

Code	Meaning / Formula	Purpose
N1	No. of students admitted in the 1st year.	Initial intake.
N2	No. of students admitted as lateral entries in 2nd year.	Added strength.
N3	No. of students readmitted / transferred from other institutes.	Additional inflow.
N4	No. of students rejoining after temporary break.	Continuation inflow.
N5	No. of students admitted through multiple entry (MEES / ABC).	Credit-based re-entries.
N6	No. of students exited through multiple exit (MEES / ABC).	Legitimate academic outflows.
Total Students Considered	N1+N2+N3+N4+N5−N6	Net batch strength for graduation calculation.

Table Format (NBA-Ready)

Year of Entry / Batch	Total No. of Students (N1 + N2 + N3 + N4 + N5 – N6)	No. of Students Graduated Within Stipulated Period (With + Without Backlogs)			
		I Year → II Year Promotion	II Year → III Year Promotion	III Year → IV Year Promotion	Graduated in 4 Years
CAY (2024-25)	–	–	–	–	– (Running Batch)
CAYm1 (2023-24)	120	118	115	112	108 (95 without / 13 with backlogs)
CAYm2 (2022-23)	115	112	110	107	103 (88 without / 15 with backlogs)
CAYm3 (2021-22)	110	108	105	102	99 (85 without / 14 with backlogs)
CAYm4 (2020-21) (LYG)	105	104	102	100	96 (80 without / 16 with backlogs)
CAYm5 (2019-20) (LYGm1)	102	100	98	95	92 (77 without / 15 with backlogs)
CAYm6 (2018-19) (LYGm2)	100	98	96	94	90 (75 without / 15 with backlogs)

4.1. Enrolment Ratio in the First Year (20)

1. Definition of Enrolment Ratio (ER)

- **Enrolment Ratio (ER) = Percentage of students actually admitted in the first year compared to the sanctioned intake.**
- **Formula (for a given academic year):**

  ER=Number of students admitted in 1st year (N1 + N3 + N4, adjusted for migrations)/Sanctioned Intake (N)×100
- **Notes:**
  - **N1 = First Year admissions (after adjusting migrations).**
  - **N3 = Separate division admissions (if any).**
  - **N4 = Supernumerary admissions (TFWS/EWS/PMSSS).**
  - **MEES entries (N5) and exits (N6) are not included in first-year enrolment.**

2. Calculation of Average ER

- **NBA does not look at a single year, but at the average enrolment ratio of the last three academic years.**
- **Formula:**

  Average ER=(ERCAY+ERCAYm1+ERCAYm2)/3

3. Points Allocation (ER → 20 Marks)

- **NBA awards up to 20 points for ER.**
- **Formula:**

  ER Points=20×Average ER/100
- **Examples:**
  - **If Average ER = 100% → ER Points = 20**
  - **If Average ER = 90% → ER Points = 18**
  - **If Average ER = 75% → ER Points = 15**
  - **If Average ER = 60% → ER Points = 12**

4. Example Calculation

Suppose Sanctioned Intake = 120 seats.

Year	Sanctioned Intake (N)	Actual Admitted (N1+N3+N4)	ER (%)
CAY (2024–25)	120	110	91.7%
CAYm1 (2023–24)	120	115	95.8%
CAYm2 (2022–23)	120	108	90.0%

- **Average ER = (91.7 + 95.8 + 90.0) ÷ 3 = 92.5%**
- **ER Points = 20 × (92.5 ÷ 100) = 18.5 ≈ 19**

5. Interpretation

- **A high ER indicates:**
  - **Strong demand for the program.**
  - **Good admission reputation and intake quality.**
- **A low ER signals:**
  - **Declining demand or poor intake quality.**
  - **Issues in placement/reputation that NBA assessors may question.**

6. Proofs/Supporting Documents

- **AICTE/University Sanctioned Intake Letters (year-wise).**
- **Admission Committee Reports / University Admission Records.**
- **Migration Approval Orders (if students moved in/out).**
- **Supernumerary Allotment Approvals (TFWS/EWS).**
- **Institution Admission Register (first year).**

7. Key Takeaways

- **Enrolment Ratio = Input quality measure under Criterion 4.**
- **Points directly proportional to Average ER across 3 years.**
- **Institutions should maintain a clear admission register showing N vs N1/N3/N4.**
- **Target → keep ER ≥ 90% to secure 18–20 points.**

Student Enrolment Ratio in the First Year under

Step-by-Step Calculation

1. Take sanctioned intake (N) for each academic year.
2. Compute N1 (regular adjusted admissions).
3. Compute N4 (supernumerary admissions).
4. Calculate ER for each year:

   $ERi=(N1+N4)/N\times100$

   where *i = CAY, CAYm1, CAYm2*.
5. Take the average across 3 years:

   $\text{Average ER}=(ER1+ER2+ER3)/3$

Example (Illustration)

Year	N (Sanctioned Intake)	N1 (Regular Admissions, adjusted)	N4 (Supernumerary)	N1+N4	ER (%)
CAY (2024–25)	120	112	5	117	97.5%
CAYm1 (2023–24)	120	115	3	118	98.3%
CAYm2 (2022–23)	120	108	4	112	93.3%

Marks distribution for the Enrolment Ratio (ER)

Concept Recap

- Enrolment Ratio (ER) = % of seats filled in 1st year admissions (adjusted for migrations + supernumerary) compared to sanctioned intake (N).
- NBA doesn't look at a single year → it takes the average ER over 3 years (CAY, CAYm1, CAYm2).
- Based on this Average ER, marks are awarded out of 20.

Marks Distribution (as per NBA Manual)

Average ER (over 3 years)	NBA Marks Awarded (out of 20)	Interpretation
≥ 90%	20 marks	Almost all sanctioned seats are filled → excellent demand for program.
≥ 80% and < 90%	17 marks	High admissions → good demand & healthy program intake.
≥ 70% and < 80%	14 marks	Decent admissions → program sustainable but needs improvement.
≥ 60% and < 70%	11 marks	Moderate admissions → possible concerns on intake quality/demand.
≥ 50% and < 60%	8 marks	Weak admissions → serious issues in reputation, placement or intake.
≥ 40% and < 50%	5 marks	Very poor admissions → program may be at risk; NBA will flag viability.
< 40%	0 marks (not mentioned, but implied)	Program has very low demand → could risk non-accreditation.

Documentation Required

- AICTE/University Approval Letters (showing sanctioned intake).
- Admission Register (first year) → signed by Admission Committee.
- Migration Records → adjustments to N1.
- Supernumerary Approvals (TFWS, EWS, PMSSS, Foreign students, PIO).

Summary Sheet showing N, N1, N4, ER% year-wise

4.2. Success Rate of the Students in the Stipulated Period of the Program (15)

1. Definition

- Success Rate (SR) measures how many students actually complete the program on time (within the stipulated duration: 4 years UG, 2 years PG, etc.), compared to how many were admitted in that batch.
- It reflects academic quality, teaching effectiveness, and student progression efficiency.

2. Formula for Success Rate (per batch)

SR=No. of students who graduated within course duration/(N1 + N2 + N3 + N5 – N6)

Where:

- N1 = Students admitted in 1st year (adjusted for migration).
- N2 = Lateral entry admissions (2nd year).
- N3 = Separate division admissions (if applicable).
- N5 = Multiple entry admissions (MEES entries).
- N6 = Multiple exit students (MEES exits).
- Numerator = Students who graduated within stipulated period (on time).

3. Average SR

- NBA takes the average of Success Rates for the last three graduating batches (LYG, LYGm1, LYGm2).

Average SR=(SRLYG+SRLYGm1+SRLYGm2)/3

4. Points Allocation

- Maximum 15 marks are awarded for SR.
- Formula:

SR Points=1.5×Average SR/10

This essentially converts the % success rate into a score out of 15.

5. Example Calculation

Suppose a B.Tech program (Sanctioned Intake = 120) →

Batch LYG (2020–21 entry)

- N1 = 110, N2 = 10, N3 = 0, N5 = 2, N6 = 2
- Denominator = 110+10+0+2−2 = 120
- Graduated on time = 108
- SR (LYG) = 108 ÷ 120 = 90%

Batch LYGm1 (2019–20 entry)

- Denominator = 118
- Graduated on time = 105
- SR (LYGm1) = 105 ÷ 118 = 89%

Batch LYGm2 (2018–19 entry)

- Denominator = 115
- Graduated on time = 102
- SR (LYGm2) = 102 ÷ 115 = 88.7%

Average SR

90+89+88.7/3=89.2%

SR Points

1.5×89.210=1.5×8.92=13.4

6. Interpretation

- **High SR (≥ 90%)** → Almost all admitted students finish on time → strong teaching-learning system.
- **Moderate SR (70–89%)** → Acceptable, but NBA may question dropout reasons (failures, transfers, poor student support).
- **Low SR (< 70%)** → Signals serious issues in retention, backlog handling, and program sustainability.

7. Documents Required for Verification

- **Admission Register** (showing N1, N2, N3, N5, N6).
- **University Result Gazette / Passing Board Reports** (graduates list).
- **Degree Award Lists** (convocation lists).
- **Backlog Clearance Records** (to confirm "on-time" graduation).
- **Migration/MEES Records** (entries and exits).

Symbol	Meaning	Formula / Definition
A	Total number of students eligible to graduate in a batch. Includes: → First-year admitted (N1) → Lateral entries (N2) → Multiple entries (N5) Minus: Multiple exits (N6).	A=N1+N2+N5−N6A = N1 + N2 + N5 - N6A=N1+N2+N5−N6
B	Number of students who actually graduated within the stipulated duration (4 years for UG).	From graduation record
SR	Success Rate of each batch	SR=B/A×100
Average SR	Mean success rate of the last three batches	Average SR=(SR1+SR2+SR3)/3

Data Table Format

Item	LYG (2020-21)	LYGm1 (2019-20)	LYGm2 (2018-19)
A = N1 + N2 + N5 – N6	132 = (120 + 11 + 1 – 0)	130 = (120 + 10 + 1 – 1)	133 = (120 + 11 + 4 – 2)
B = Graduated within 4 years	123	121	124
Success Rate (SR = B/A × 100)	123/132×100=93.18%	121/130×100=93.07%	124/133×100=93.23%
Average SR of 3 batches			(93.18+93.07+93.23)/3=93.16%

Supporting Documents to Maintain

Evidence	Source / Section Responsible
Admission register showing N1, N2, N5, N6	Admission Cell / Registrar
Graduation list with roll numbers & pass status	Controller of Examinations
Lateral-entry & MEES approvals	University / DTE / ABC portal
IQAC attested “Success Rate Summary Sheet”	Department & IQAC

4.3. Academic Performance of the First-Year Students of the Program (10)

- **Overview**
  - **The academic performance of first-year students is a significant indicator of the quality of intake, effectiveness of teaching-learning practices, and the institutional support ecosystem.**
  - **Monitoring and improving first-year results is critical, as it sets the foundation for progression through higher semesters and impacts retention and success rates.**
- **Definition & Calculation**
  - **Academic Performance Index (API) for first-year students is computed as:**
    - **API = X × (Y/Z)**
      - **X = Mean of first-year Grade Point Average (GPA) on a 10-point scale or (Mean percentage of marks ÷ 10).**
      - **Y = Total number of successful students (students who passed all prescribed first-year courses).**
      - **Z = Total number of students who appeared in the first-year examinations.**
  - **Average API is calculated over three consecutive years (CAYm1, CAYm2, CAYm3).**
  - **This ensures uniformity in performance reporting across batches.**

- **Evidence & Documentation**
  - **Year-wise first-year result analysis reports (pass percentage, API, topper details, GPA distribution).**
  - **Consolidated API calculation sheet for CAYm1, CAYm2, and CAYm3.**
  - **Records of remedial classes, bridge courses, and student mentoring logs.**
  - **Examination result gazette copies verified by the affiliating university.**
- **Strengths**
  - **Strong first-year foundation due to effective teaching-learning practices.**
  - **Structured mentoring and remedial measures improve pass rates.**
  - **Consistent API performance across three consecutive batches.**
- **Areas of Improvement**
  - **Additional academic support for students admitted through lateral entry or diverse educational backgrounds.**
  - **Enhanced focus on practical skill integration in the first year.**

    More systematic tracking of slow learners with targeted improvement plans.

First-Year Academic Performance

- CAYm1: Compute $API_1 = X_1 \times (Y_1/Z_1)$.
- CAYm2: Compute $API_2 = X_2 \times (Y_2/Z_2)$.
- CAYm3: Compute $API_3 = X_3 \times (Y_3/Z_3)$.
- Average API (First Year) = $(API_1 + API_2 + API_3) \div 3$.

First-Year Data

Cohort	Z (Appeared)	Y (Successful)	X (Mean GPA or %/10)	API = X × (Y/Z)
CAYm1	120	100	6.8	6.8 × (100/120) = 5.67
CAYm2	118	105	7.2	7.2 × (105/118) = 6.41
CAYm3	115	95	6.9	6.9 × (95/115) = 5.70

Supporting Documents

- Result Gazette / Mark sheets → compute mean GPA/% of successful students (X).
- Promotion / Pass lists → confirm successful student counts (Y).
- Exam appearance lists → confirm total appeared (Z).
- Computation sheet → show how X, Y/Z, and API were derived (for audit transparency).

4.4 Academic Performance of the Second Year Students of the Program (10)

1. Objective

- NBA wants to evaluate not just first-year performance but also how students sustain their performance in the second year.
- The idea is to see whether students are learning well enough to progress smoothly to the 3rd year.
- This is measured through the Academic Performance Index (API) of second-year students.

2. The API Formula

API=X×(Y/Z)

- X (Quality factor)
  - *Option A (preferred):* Mean GPA of all successful 2nd-year students (on a 10-point scale).
  - *Option B:* Mean % marks of successful 2nd-year students ÷ 10.
  - Only successful students are included in this calculation.
- Y (Successful students)
  - Total number of students who passed the 2nd year and were promoted to the 3rd year.
  - Source: Official Pass/Promotion List issued by University/Autonomous Body.
- Z (Appeared students)
  - Total number of students who actually appeared in the 2nd-year exams (both pass + fail).
  - Absentees, detained, debarred are excluded.
  - Source: Exam Appearance List/Nominal Roll.

3. Steps to Calculate

Collect Z → From Exam Attendance/Appearance List → count Present students.
Collect Y → From Pass List → count students who were promoted to 3rd year.
Calculate X → From Gazette → take GPA/marks of successful students (those in Y), compute their mean.
Plug into formula → API=X×(Y/Z)
Do this for three consecutive cohorts → CAYm1, CAYm2, CAYm3.
Average them →

Average API=AP1+AP2+AP3/3

4. Worked Example

Suppose we have 3 cohorts (CAYm1 = 2023-24, CAYm2 = 2022-23, CAYm3 = 2021-22):

CAYm1 (2023–24)

- Z (Appeared) = 110 students
- Y (Successful) = 95 students promoted to 3rd year
- Mean GPA of successful = 7.2
- API = 7.2 × (95 ÷ 110) = 7.2 × 0.864 = **6.21**

CAYm2 (2022–23)

- Z = 108
- Y = 92
- Mean GPA = 6.8
- API = 6.8 × (92 ÷ 108) = 6.8 × 0.852 = **5.79**

CAYm3 (2021–22)

- Z = 112
- Y = 96
- Mean GPA = 7.0
- API = 7.0 × (96 ÷ 112) = 7.0 × 0.857 = **6.00**

Average API

(6.21+5.79+6.00)/3=6.00(out of 10

Table Layout (for SAR)

Metric	CAYm1	CAYm2	CAYm3
X (Mean GPA or %/10 of successful)	7.2	6.8	7.0
Y (Successful students)	95	92	96
Z (Appeared students)	110	108	112
API (X × Y/Z)	6.21	5.79	6.00
Average API	colspan=3 → 6.00		

Data Sources Required

For audit & SAR submission, you must maintain:

- **Second-Year Result Gazette → student-wise marks/GPA (to compute X).**
- **Pass/Promotion List → confirms Y (successful).**
- **Exam Appearance List → confirms Z (appeared).**
- **Computation Sheet (Excel) → showing transparent calculation of X, Y/Z, API.**

4.5 Academic Performance of the Third Year Students of the Program (10)

1. Purpose

- NBA requires evaluating academic consistency across years – 1st, 2nd, and 3rd year.
- This indicator checks how well students perform in 3rd year, and how many progress successfully to the final year (4th year).
- It balances academic quality (GPA/marks) with progression efficiency (pass rate).

2. Formula

API=X×(Y/Z)

Where:

- X (Quality Factor):
  - *Option A (preferred):* Mean GPA of all successful 3rd-year students (10-point scale).
  - *Option B:* Mean % marks of successful 3rd-year students ÷ 10.
  - Note: *Only successful students are included in the mean*.
- Y (Successful Students):
  - Students who cleared 3rd year and are promoted to the 4th year (final year).
  - Source: Official Pass/Promotion List.
- Z (Appeared Students):
  - Students who appeared in 3rd year exams (Present in Exam Attendance List, Pass + Fail).
  - Excludes Absent, Detained, Debarred.
- Successful students = Those eligible for 4th year progression.

3. Calculation Steps

Collect Z → from Exam Appearance List (count Present = Z).
Collect Y → from Pass/Promotion List (count promoted to 4th year).
Calculate X → from Result Gazette (compute mean GPA or %/10 of only the Y students).
Compute API → apply formula per cohort (CAYm1, CAYm2, CAYm3).
Average API →

Average API=APICAYm1+APICAYm2+APICAYm3/3

4. Worked Example

Cohort CAYm1 (2023–24):

- **Z = 105** appeared.
- **Y = 92** promoted to 4th year.
- **X = 7.1** (mean GPA of successful).
- **API = 7.1 × (92/105) = 7.1 × 0.876 = 6.22**

Cohort CAYm2 (2022–23):

- Z = 108, Y = 90, Mean GPA = 6.9.
- API = 6.9 × (90/108) = 6.9 × 0.833 = **5.75**.

Cohort CAYm3 (2021–22):

- Z = 110, Y = 94, Mean GPA = 7.0.
- API = 7.0 × (94/110) = 7.0 × 0.855 = **5.99**.

Average API

(6.22+5.75+5.99)/3=6.00(out of 10)

5. Interpretation

- **High API (≥ 8/10):** Strong performance and smooth progression to final year.
- **Medium API (5–7.9):** Acceptable, but indicates some dropouts/failures.
- **Low API (< 5/10):** Weak performance; signals academic or support issues.

6. Supporting Documents (Mandatory Proofs)

- **3rd Year Result Gazette** → for GPA/marks (to compute X).
- **Pass/Promotion List** → confirms Y (successful → promoted to 4th year).
- **Exam Appearance List** → confirms Z (appeared).
- **Mapping sheet** → roll number-wise linkage across Gazette, Pass List, and Attendance.

Suggested SAR Table Format

Metric	CAYm1	CAYm2	CAYm3
X (Mean GPA or %/10 of successful)	7.1	6.9	7.0
Y (Successful → promoted to 4th year)	92	90	94
Z (Appeared in 3rd-year exam)	105	108	110
API = X × (Y/Z)	6.22	5.75	5.99
Average API (out of 10)	–	–	6.00

4.6 Placement, Higher Studies and Entrepreneurship (30)

1. Purpose

- NBA evaluates not just students' performance during the program, but also outcomes after graduation.
- This sub-criterion measures how many graduates get:
  Placed in jobs (campus/industry recruitment).
  Admitted to higher studies (PG, research, foreign universities, etc.).
  Startups / Entrepreneurship.
- The combined metric is expressed as a Placement Index (P), and the marks are scaled accordingly.

2. Formula

Placement Index Points=0.3×Average Placement Index (P)

Where:

- P (Placement Index) is calculated year-wise for the last three batches:
  P={Number of students placed + Number of students for higher studies + Number of students as entrepreneurs}/Numbe r of students graduated×100
- Average Placement Index (P)
  Pavg=PCAY+PCAYm1+PCAYm2/3
- Placement Index Points (out of 30)
  =0.3×Pavg= 0.3 \times P_{avg}=0.3×Pavg

(So if average P = 80%, then Points = 0.3 × 80 = 24 out of 30)

3. Definitions

- Graduated students = Students who completed the program in stipulated time (LYG batch).
- Placed students = Students employed through campus placements or outside offers with proof.
- Higher studies = Students enrolled in PG programs (M.Tech, MS, MBA, PhD, etc.) with admission proof.
- Entrepreneurs = Students who started startups/business ventures with proper registration evidence.
- Excluded = Students who:
  - Opted out of placement voluntarily.
  - Did not appear for exams or did not graduate.

4. Step-by-Step Computation

Collect total graduated students (denominator) → from University Degree Award List.

Count placed students (with offer letters, joining reports).

Count higher studies students (with admission letters).

Count entrepreneurs (with company incorporation/registration proof).

Compute P for each cohort →

$$P = \text{Placed} + \text{Higher Studies} + \text{Entrepreneurs} / \text{Graduates} \times 100$$

Compute average P across 3 years (CAY, CAYm1, CAYm2).

Compute Points = 0.3 × Pavg (max 30).

5. Example Calculation

Batch	Graduated (G)	Placed	Higher Studies	Entrepreneurs	P (%) = (Placed+HS+Ent)/G × 100
CAY (2024)	120	72	18	5	(72+18+5)/120 ×100 = 79.2%
CAYm1 (2023)	115	70	15	3	(70+15+3)/115 ×100 = 76.5%
CAYm2 (2022)	110	65	20	2	(65+20+2)/110 ×100 = 78.2%

- **Average P** = (79.2 + 76.5 + 78.2)/3 = **78.0%**
- **Placement Index Points** = 0.3 × 78.0 = **23.4 ≈ 23/30**

Supporting Documents Required

- Placement records: offer letters, joining reports, company HR confirmation.
- Higher studies records: admission letters, enrollment certificates.
- Entrepreneurship proofs: company registration, GST/Startup India certificate.
- Graduation list: University Degree Award List to confirm denominator.
- Summary sheets: student-wise mapping with status.

Common Pitfalls

- Counting students with multiple offers more than once → count only once.
- Counting students without proof of placement/admission → not allowed.
- Including students who did not graduate in denominator → wrong.
- Ignoring entrepreneurs due to lack of formal evidence → ensure proper documentation.

Table Format for SAR

Metric	CAY	CAYm1	CAYm2
Graduates (G)	120	115	110
Placed	72	70	65
Higher Studies	18	15	20
Entrepreneurs	5	3	2
Placement Index P (%)	79.2	76.5	78.2
Average P (%)	colspan=3 → 78.0%		
Placement Index Points (out of 30)	colspan=3 → 23.4 ≈ 23		

4.7 Professional Activities (25)

4.7.1 Professional Societies/ Bodies, Chapters, Clubs, and Professional Engineering Events Organized (05)

- **1. Purpose of this Criterion**
  - **To evaluate how actively the program engages students and faculty in professional societies and clubs, both at institutional and external levels.**
  - **Measures the extent of exposure students get to professional networking, engineering practices, and technical events.**
  - **Encourages holistic development of students beyond academics.**
- **2. Sub-components (as per NBA framework – typically 5 points)**

a) Professional Society Memberships

- **Institution/Department must establish student chapters of recognized national/international professional societies such as:**
  - **IEEE, ACM, ASME, SAE, ISTE, IETE, CSI, IEI, Institution of Engineers, SAE, ISHRAE, etc.**
- **Students should be enrolled members, with at least 20–30% of total strength participating.**
- **Faculty coordinators must actively guide and mentor student chapters.**
- ***Evidence:* Membership certificates, lists of registered students/faculty.**

b) Student Chapters and Clubs

- **Active operation of departmental clubs (Coding Club, Robotics Club, AI-ML Club, Innovation Club, Green Energy Club, etc.).**
- **Clubs conduct activities like guest lectures, workshops, hackathons, design competitions, quizzes.**
- **Such clubs enhance peer-to-peer learning, innovation culture, and leadership qualities.**
- ***Evidence:* Event reports, posters, attendance, photographs.**

c) Professional Engineering Events Organized

- Institution should organize technical events under the banner of these societies/clubs:
  - Conferences, seminars, guest lectures, industrial conclaves, design contests, technical fests.
  - National/International-level events are given more weight.
  - Student participation in external society events (paper presentations, competitions) adds credibility.
- *Evidence:* Event reports with brochures, schedules, attendance, photos, certificates.

S.N.	Name of the Professional Society/Body/Chapter/Club	Type (Faculty/Student/Joint)	Year of Establishment	Membership Strength	Faculty Coordinator	Major Activities Organized (Last 3 Years)
1	IEEE Student Chapter	Student Chapter	2018	120 Students, 5 Faculty	Dr. A. Kumar	Technical paper contests, IEEE Day, Hackathon
2	ACM Student Chapter	Student Chapter	2019	90 Students, 3 Faculty	Prof. S. Mehta	Coding competitions, Research seminars, Workshop on AI
3	ISTE Faculty & Student Chapter	Joint Faculty-Student	2017	200 (faculty + students)	Dr. R. Sharma	Faculty Development Programs (FDPs), Student Symposium
4	CSI Student Chapter	Student Chapter	2020	80 Students	Prof. N. Gupta	Web development bootcamps, App design contest
5	Robotics & Automation Club	Departmental Club	2016	60 Students	Prof. V. Singh	Robotics competition, Drone workshop, National Robotics Day
6	Green Energy & Sustainability Club	Departmental Club	2021	45 Students	Prof. M. Reddy	Solar energy awareness drive, Green hackathon
7	Entrepreneurship & Innovation Cell (EIC)	Institution-Level Club	2015	100 Students, 10 Alumni Mentors	Dr. P. Das	Startup Pitch Fest, Incubation workshops, IPR Seminar
8	Institution of Engineers (India) – IEI Student Chapter	Student Chapter	2018	75 Students	Prof. K. Patel	Bridge Design Contest, IEI Technical Quiz

List of Events / Programs Organized

S.N.	Name of the Professional Society / Body / Chapter / Club	Name of the Event	National / International Level	Date of Event	Academic Year (CAYm1 / CAYm2 / CAYm3)
1	IEEE Student Chapter	Hackathon 2023	National	14-10-2023	CAYm1
2	ACM Student Chapter	Research Seminar on AI	International	20-03-2023	CAYm1
3	ISTE Student Chapter	Faculty Development Program	National	10-12-2022	CAYm2
4	CSI Student Chapter	Coding Competition	National	15-09-2022	CAYm2
5	Robotics & Automation Club	Robotics Design Contest	National	28-01-2021	CAYm3
6	Green Energy & Sustainability Club	Renewable Energy Workshop	International	18-06-2021	CAYm3

- **Evidence & Documentation Required for NBA**
  - List of professional society memberships with student/faculty counts.
  - Records of student chapters/clubs with activity reports.
  - Event calendars, brochures, and reports of professional engineering events organized.
  - Attendance sheets, photos, and external participation certificates.
  - Faculty coordinator reports verifying chapter activities.

4.7.2 Student’s Participations in Professional Events (10)

1. Purpose of this Criterion

- To assess how actively students are involved in external professional activities beyond their home institution.
- Demonstrates exposure to real-world competitions, innovation platforms, and peer benchmarking.
- Reflects institutional support in motivating and enabling students to showcase skills in hackathons, ideathons, conferences, seminars, workshops, and competitions.

2. Definition of Professional Events

- Events conducted by other institutes, industries, or professional societies at state, national, or international level.
- Examples include:
  - Hackathons / Codeathons (e.g., Smart India Hackathon, SIH; ACM ICPC; Kaggle competitions).
  - Ideathons / Start-up Challenges (e.g., NASSCOM challenges, incubation contests).
  - Technical Paper Presentations / Seminars / Conferences.
  - Project Competitions / Robotics Competitions / Quiz Contests.
  - Workshops, Symposiums, FDPs, and technical exhibitions.

3. Data to be Reported (3-year basis: CAYm1, CAYm2, CAYm3)

For each student participant, details should include:

- Name of the Student.
- Program & Year of Study.
- Name of the Event (Hackathon, Ideathon, Paper Presentation, etc.).
- Organizing Institute / Body.
- Level of Event (State / National / International).
- Date of Event.
- Outcome/Achievement (Participation, Winner, Runner-up, Finalist, Certificate of Merit).

List of Students Participated in Professional Events (Last 3 Years)

S.N.	Name of the Student	Name of the Event	Level (State / National / International)	Date of Event	Name of the Award (if any)	Academic Year (CAYm1 / CAYm2 / CAYm3)
1	John Doe	Smart India Hackathon 2023	National	12-08-2023	Finalist	CAYm1
2	Priya Sharma	ACM ICPC Regionals	International	22-11-2022	–	CAYm1
3	Rahul Nair	Robotics Competition	National	15-04-2022	Winner	CAYm2
4	Meera Iyer	Green Energy Ideathon	State	10-10-2021	Runner-up	CAYm3
5	Akash Verma	Coding Challenge	National	08-09-2021	Participation	CAYm3

. Evidence & Documentation for NBA

- **Year-wise list of student participations (with event details).**
- **Participation certificates, merit/winner certificates.**
- **Event brochures/posters with student names listed.**
- **Institutional support letters (travel grants, sponsorships).**
- **Photographs, media coverage, or online event references.**

4.7.4 Student Publications (05)

- **Overview**
  - Student publications are an important measure of the research culture, innovation capacity, and academic engagement of the department.
  - Encouraging students to publish in reputed journals and conferences provides them with early exposure to scholarly writing, peer review processes, and global research platforms.
  - The institution provides structured mentoring and support to ensure that students actively contribute to research and innovation.
- **Definition and Scope**
  - Student publications considered under this metric include:
    - Research papers in peer-reviewed journals (National/International).
    - Articles in conference proceedings with ISBN/DOI.
    - Contributions to book chapters or indexed publications.
  - Publications must have student(s) as authors/co-authors; faculty guidance and joint authorship are encouraged.
  - Records are maintained for the last three consecutive years (CAYm1, CAYm2, CAYm3).
- **Institutional Efforts to Promote Student Publications**
  - Establishment of a Research & Development Cell and Student Research Forum.
  - Mentorship by faculty for technical writing, data analysis, and publication ethics.
  - Workshops on research methodology, plagiarism awareness, and citation practices.
  - Financial support for registration and publication charges in reputed journals/conferences.
  - Encouragement for students to convert final-year projects and mini-projects into research papers.

Sample Annexure Table Format

S.N.	Title of Paper/Article	Name of Student(s)	Journal/Conference Name	Publisher/Organizer	ISSN/ISBN/DOI	Indexed In	Level (National/International)	Year (CAYm1/CAYm2/CAYm3)	Recognition/ Award
1	AI for Smart Cities	Priya Sharma (CSE, VII Sem)	IEEE Conference on AI	IEEE	DOI: 10.1109/xyz123	Scopus	International	CAYm1 (2023)	Best Paper Award
2	Renewable Energy Models	Rahul Nair (ME, V Sem)	Int. Journal of Green Tech	Springer	ISSN: 2395-1111	Web of Science	International	CAYm2 (2022)	–
3	IoT in Healthcare	Ananya Gupta (ECE, VI Sem)	National Conference on IoT	NIT Delhi	ISBN: 978-93-555	–	National	CAYm3 (2021)	–

- **Evidence and Documentation**
  - Year-wise list of student publications with details of authors, journal/conference name, publisher, ISSN/ISBN/DOI, indexing, and awards.
  - Copies of certificates, acceptance letters, published papers, and proceedings.
  - Department maintains archives of student publications both in softcopy (institutional repository) and hardcopy (department library).